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Exploiting Privacy Preserving Prompt Techniques for Online Large Language Model Usage

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Abstract

Online Large Language Models (LLMs) are widely employed across various tasks, including privacy-sensitive ones like financial advice or paragraph rewriting. Presently, users directly submit prompts to online LLM servers, inadvertently revealing sensitive keywords and facilitating server tracking to build user profiles. In this paper, we propose a local privacy-preserving prompt assistant (LPPA) that provides users with a usable method to balance privacy in the prompts and the utility of the LLM output. The LPPA will analyze the users' prompts, suggest modifying the prompts to protect the sensitive keywords, and provide an inference of the potential utility impact of the online LLM output. Specifically, we first propose a privacy module to identify the sensitive keywords in the prompt and adopt four privacy techniques, including remove, mask, replace, and rewrite to hide the keywords. While these techniques affect the utility of online LLM output, we measure such impact using the LLM output of the original prompt and modified prompts and discuss the cases with high, median, and low impact. In addition, we propose a utility inference model to infer the utility impact locally without disclosing the prompts to the online LLM. We evaluated LPPA on the real-world users' prompts and showed that the remove technique achieves the best performance, and it empowers users with meaningful ways to adjust their prompts to safeguard their privacy while still maintaining a satisfactory level of utility in online LLM usage.

Index Terms—

Privacy-preserving techniques; large language models; online and offline

I. Introduction

In recent years, the proliferation of Large Language Models (LLMs) has transformed the landscape of natural language processing, offering unparalleled multi-task capabilities such as rewriting, translation, advising, and code generation. As of March 2024, the best-performed LLMs are primarily proprietary online platforms accessible solely through web interfaces or APIs, e.g., ChatGPT [1]. However, the widespread adoption of these online

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LLMs has raised significant concerns about the continuous collection of users' privacy-sensitive information. Previous research has demonstrated vulnerabilities in the inference stage of LLMs, highlighting the potential for Personally Identified Information (PII) leakage through carefully crafted prompts [2]–[4]. Moreover, users' prompts sent to the online LLMs pose additional privacy risks, as these prompts may contain privacy-sensitive keywords. The disclosure of privacy-sensitive keywords can produce unexpected harm to users, e.g., target advertising and profile inference, while the privacy-sensitive keywords may not necessarily be needed to accomplish users' intention of using online LLMs. Given the potential implications, it is imperative to address privacy protection concerns and develop a local assistant to assist users in handling privacy-sensitive keywords in their prompts.

Previous works protect the privacy of users' prompts within limited tasks, ignoring the diversity of the real user prompts sent to online LLMs. Specifically, Lin et al. [5] used Emoji to encrypt the users' prompts, and their work can only apply to classification tasks; Chen et al. [6] proposed a hide-and-seek model that can apply to the classification, summarization, and translation tasks. However, state-of-the-art LLM can perform many more tasks where user prompts are diverse in style. In LMSYS-CHAT-1M [7], the authors collected one million real-world user prompts and clustered them into 20 clusters, e.g., discussing software errors, inquiries, creating and improving business strategies, code generation. In Wildchat dataset [8], the authors found even more diverse prompts than LMSYS-CHAT-1M by t-SNE visualization. We found in the Wildchat dataset that 59.67 % of user prompts may contain privacy-sensitive keywords. Hiding the privacy-sensitive keywords in these prompts may or may not lead to a high utility impact on the online LLM output. For example, in the prompt "Suggest a travel plan to Boston," removing "Boston" leads to a high utility impact; in the prompt "Rewrite I love Boston because it is a big city," removing "Boston" leads to a low utility impact. In the diverse task setting, a local privacy-preserving prompt assistant is needed to inform users of the privacy gain and utility impact when hiding the sensitive keywords in the prompts.

This paper proposes a local privacy-preserving prompt assistant (LPPA) to address privacy issues in online LLM usage. It contains three modules: privacy module, analysis module, and user-interface module. The privacy module suggests possible privacy-sensitive keywords in the user prompts. Users may accept or reject the privacy suggestions or propose more keywords that are not in the suggestions. Then, users could select a privacy technique to hide the selected keywords and obtain modified prompts. After that, the analysis module takes the original and modified prompts as input and infers the privacy gain and the utility impact. The user-interface component visualizes the above result, and users decide whether or not to use the modified prompt. Our contributions are three-fold:

First, we demonstrate the potential utility impact of hiding sensitive keywords in user prompts using four techniques: remove, mask, replace, and rewrite. We found the remove technique resulted in the lowest utility impact, with 63.57% of prompts having a low impact and 1.36% having a high impact. We also found the rewrite technique resulted in the highest utility impact, with 33.81% of prompts having a low impact and 3.49% having a high impact.

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Second, we propose a utility inference model to infer the utility impact. The model is trained using the WildChat dataset and implemented locally so that the original prompt is not disclosed to the online LLMs. Results show the model achieves 71% and 64% in Spearman and Pearson correlations on the utility inference, demonstrating the effectiveness of our model.

Third, we integrated the privacy and analysis modules with efficient and intuitive user-interface modules for a complete LPPA system. LPPA presents the results of the analysis of privacy techniques in a meaningful way to assist users in selecting modified prompts.

II. Related work

Privacy in LLM training.

The advent of Large Language Models (LLMs) has escalated privacy concerns by risking the exposure of sensitive information [9]–[11]. Research by Carlini et al. [12], Kim et al. [3], and Inan et al. [4] reveals that LLMs can unintentionally memorize and leak PII during inference, underscoring the critical demand for robust privacy safeguards. Researchers have implemented various privacy-enhancing techniques during the training and inference phases to mitigate the risks of privacy and PII exposure. Methods such as differential privacy [13], federated learning [14], and encrypted computation [15] have become crucial for minimizing the exposure of sensitive information.

Privacy of LLM prompts.

Despite advancements in protecting privacy during the training and inference processes, the issue of sensitive information in user-generated prompts has received less attention. Researchers have proposed prompt sanitization and anonymization techniques. While innovative, the Hide and Seek (HaS) [6] framework encounters several constraints in offering full-scale privacy safeguards. Primarily, its mechanism is tailored to obscure named entities, leaving verbs, non-named entity terms, and the architecture of sentences exposed and potentially compromising other categories of sensitive information. Yim et al. [16] proposed to hash personal data entities using a hash function, thereby sending the hash values to the remote LLM. The hash values in the LLM's response are then converted back to their original entities. This approach allows the use of personal data in LLM prompts without exposing sensitive information. EmojiCrypt [5] proposes a new prompt encryption method that encrypts sensitive user information in prompts using a sequence of emojis and operators. While it has proven the method's practicality and its ability to maintain LLM performance on real datasets, the limited variety of symbols and operators could lead to information loss. Similarly, the VPASS framework [17] utilizes deep transfer learning to accurately identify privacy-sensitive prompts based on users' historical interaction records with the Voice Assistant System (VAS) [18], thereby reducing the risk of disclosing sensitive information to VAS. In this paper, we propose a novel system, LPPA, to assist users in balancing privacy with utility by suggesting alternative prompts.

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III. System model

LPPA is a local system running on a user's device. The user's original prompt stays locally, while the modified prompts are sent to the online LLMs. The modified prompts do not contain selected keywords, thus preserving privacy. However, using the modified prompt may result in a different online LLM output. To help the user choose the modified prompt, LPPA generates a set of metrics associated with each prompt, including privacy techniques, privacy gain, and utility impact. As shown in Figure 1, LPPA consists of a privacy module, an analysis module, and a user-interface module.

The **privacy module** aims to generate multiple modified prompts based on the user's original prompt using different modification techniques. Denote the user's original prompt as I_u, the modified prompts as $\{I_i | 1 \leq i \leq n\}$, and the privacy module as $f_p(I_u) = \{I_i | 1 \leq i \leq n\}$.

The **analysis module** aims to generate a set of metrics $\{m_{i,j} | 1 \leq j \leq k\}$ associated with a modified prompt I_i. The analysis module is denoted as $f_a(I_i, I_u) = \{m_{i,j} | 1 \leq j \leq k\}$.

The **user-interface** module aims to provide an intuitive overview of modified prompts $\{I_i | 1 \leq i \leq n\}$ and their associated metrics $\{m_{i,j} | 1 \leq i \leq n, 1 \leq j \leq k\}$. For example, the modified prompts can be ranked based on either the privacy gain or the utility impact. Eventually, the user selects a satisfactory prompt to submit it to the online LLMs.

IV. LPPA implementation

This section introduces the detailed implementation of the privacy, analysis, and user-interface modules.

A. Privacy module

We propose two steps to generate modified prompts from the user's original prompt, as shown in Fig. 2. First, we used a set of 54 pre-defined privacy-sensitive categories $\{C_i | 1 \leq i \leq 54\}$, including name, city, job type, etc.[1] Then, according to these categories, we find sensitive keywords in the user's original prompt I_u via a **name-entity-recognition model**. The keywords in category C_i are denoted as $W_{u,i}$. Second, we hide the keywords via four modification techniques. In this paper, we mainly focus on the one-keyword case to demonstrate the concept. In the following, we use the original prompt "why do sales engineering managers like their job?" and the keyword "sales engineering" as an example.

1. Remove a word w from the user input. The modified prompt is "why do like their job?"
2. Mask a word w with its category name c. The modified prompt is "why do [jobtitle] like their job?"
3. Replace words w with another word in the same category c. The modified prompt is "why do Regional Brand Analysts like their job?"

[1] https://huggingface.co/datasets/ai4privacy/pii-masking-200k

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4. Rewrite the input I_u using a local LLM with the instruction "Rewrite I_u removing w," and the modified prompt is "Why do engineering managers like their job:"

For the prompt, the privacy module finds n sensitive keywords. We apply the above four techniques to each keyword. Finally, the privacy module outputs $4n$ modified prompts.

B. Analysis module

For each modified prompt, the analysis module aims to analyze the original prompt I_u and the modified prompt I_i together from three perspectives: prompt difference, privacy gain, and utility impact, as shown in Figure 3.

Prompt Differences. Showing the prompt differences is an efficient and intuitive way for users to understand LPPA's modified prompts. It is important for users to gain confidence in privacy because the modified prompt does not contain the selected keyword, and it is also important to gain confidence in the relevance of the LLM output of the modified prompt due to the minimum changes. We analyze the differences using the following metrics: the number of occurrences in the original and the semantic similarity between the two prompts. An example of the prompt differences is shown in Table I.

Privacy gain. To help users understand the privacy gain of each privacy technique, we provide the analysis as follows.

- Remove: the keyword is removed from the user's original prompt. The modified prompt does not add any additional information to the original prompt. Remove technique achieves the highest privacy gain.
- Mask: the keyword is replaced with its category name. Thus, the keyword's category is disclosed to the online LLM instead of a specific keyword. Using the category in the prompt mitigates the damage to the semantic meaning in the hope of minimizing the impact on the utility.
- Replace: the keyword is replaced by a random keyword in the same category. The replace technique is more aggressive than the remove technique because the random word is added as new information. This technique effectively disables online LLMs from tracing an accurate user profile. However, this technique may have some unexpected utility impacts due to the randomness.
- Rewrite: a local LLM is used to rewrite the original prompt. This technique completely relies on the local LLM, which may or may not produce a better result. The local LLM may find a prompt to achieve both higher privacy and higher utility than remove, mask and replace techniques. However, it may significantly change the prompt to cause a negative impact on the utility, or it may replace the keyword with a different but semantically similar keyword, which is not desired by the user.

Utility impact. LPPA integrates a customized model to infer the utility impact accurately. As shown in Figure 3, the utility impact is derived from the user original prompt I_u and the modified prompt I_i. Therefore, we train a local utility inference model using

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an existing dataset. The model training does not use the online LLM outputs as input because, in the use case, the inference has to be done without the knowledge of online LLM outputs. We develop a local utility inference model with three steps, as shown in Figure 4. We first construct a dataset with a pair of original and modified prompts using four privacy-preserving techniques and then query the LLMs to obtain the corresponding original and modified output. Secondly, we obtain the utility impact labels. We define the utility impact as the semantic similarity between the two outputs corresponding to the two prompts. Specifically, we use cosine semantic similarity as the labels. The range of the similarity score is [−1, 1]. We consider the similarity score [0.8, 1] as low impact, (0.3, 0.8) as median impact and [−1, 0.3] as high impact. According to the above definition, we obtained the utility impact label using the embedding vectors of the LLM outputs via a pre-trained embedding model (all-mpnet-base-v2). The embedding model takes a text pair as input, calculates an embedding pair (u, v) for the text pair, and then outputs a single cosine similarity value [19]. Thirdly, we fine-tuned the pre-trained embedding model using the original prompt and the modified prompt as input and the semantic similarity label as the output. We used the cosine similarity loss to fine-tune the embedding model. We consider fine-tuning could align the model embedding to measure the semantic similarity from the prompt input space to the LLM output space. This way, our utility inference model can infer the semantic similarity of the outputs, which will be used as the utility impact.

C. User-interface module

The user-interface module presents the analysis module's results to the users, including general and prompt results.

General results include results of all prompts per privacy techniques: the semantic similarity of two prompts and the utility impact, i.e., the semantic similarity of two LLM outputs. These results help users to understand the privacy gain and utility impact when choosing different privacy techniques, as shown in Figures 5(a) and (b).

Prompt results are related to the user's current prompt. LPPA uses the privacy module to generate modified prompts and uses the analysis module to generate results associated with all modified prompts, including prompt difference (Table I), privacy gain (Section IV-B), and utility impact. For the utility impact, LPPA presents the distribution of the inference results for the inferred utility impact. We show an example of the inferred utility impact 0.7, highlighted in Figure 6, where users can easily understand the probabilities and the errors of the inaccurate inferences on a specific inferred utility impact.

V. Evaluation

In this section, we describe the dataset and evaluation details and then present the results of the utility impact of privacy techniques and the utility inference model.

A. Dataset and implementation

We used WildChat dataset [8], which contains 570K real user prompts with online LLM ChatGPT. We focused on the English subset (prompts with ASCII characters) and considered the first prompt in the conversations for simplicity. The total number of such

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prompts is 215K. Because not all prompts contain sensitive keywords, we used a pre-trained NER model to find the sensitive keywords within the pre-defined 54 sensitive categories, e.g., name, city, job type[2]. We kept the prompts that have at least a keyword with an NER score ≥ 0.7 and obtained a dataset of 96793 prompts with sensitive keywords. For these prompts, we made pairs of each prompt and each keyword (NER scores ≥ 0.7); one prompt can be paired with multiple keywords, and one keyword can be paired with multiple prompts. Finally, we selected the top 200 pairs of the highest NER scores in each category to ensure the sensitive keywords are likely to be identified accurately and all sensitive categories are covered. As the number of pairs of keywords in some categories is less than 200, this process yielded a total of 7623 pairs (≤ 10800 = 200*54). We applied the remove, replace, mask, and rewrite techniques to these prompts and keywords. Note that if a keyword appears multiple times, we process all occurrences in the modified prompts. We applied the Llama 2 7B model [20] to generate the original and modified LLM output. We make an 80%/20% split for training and evaluating the utility inference model.

B. Performance analysis

1) Impact vs. privacy techniques: We show the impact of prompt similarity and LLM output similarity vs. privacy techniques in Figure 5. We consider a privacy technique to be effective if it causes a low impact on most prompts and outputs and to be ineffective if it causes a high impact on most prompts and outputs. We found that among the four techniques, the remove technique is the most effective as the numbers of lowly impacted prompts and outputs are the most (94.74%, 63.57%), and the numbers of highly impacted prompts and outputs are the least (0.03%, 1.37%). The remove technique is recommended as an effective technique. On the other hand, the rewrite strategy is an ineffective technique, as the numbers of lowly impacted prompts and outputs are the least (29.33%, 33.81%), and the numbers of highly impacted prompts and outputs are the most (1.42%, 3.49%). In the rewrite technique, we found the local LLM model may have confused the rewrite instruction with the instruction in the original prompt. The model may partially execute the instruction in the original prompt, leading to a significant utility change in both the prompt and the output. The rewrite technique is not recommended but may have room to improve with the advancement of LLM and instructions. The replace and mask techniques are slightly worse than the remove technique.

2) Utility impact vs. keyword category: We analyzed the utility impact on different categories. Since the remove strategy achieved the best performance, we analyze the remove technique in detail. We found low-impacted categories include currency symbol, amount, date, last name, time, etc. Removing keywords in these categories in the LLM prompt, e.g., “$” (currency symbol), has limited impacts on the LLM output. On the other hand, high-impacted categories include URL, job title, city, county, etc., and the keywords of these categories are likely to be the key components of the prompts, e.g., the keyword “System Analysis Engineer” in the prompt “Job advertisement on LinkedIn to request a System Analysis Engineer.” Removing them from the prompt may significantly impact the utility of the LLM output.

[2] https://huggingface.co/Isotonic/distilbert_finetuned_ai4privacy_v2

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3) Utility impact vs prompt topic: We clustered the 7623 prompts into 20 clusters using the k-means algorithm and used GPT-4 to summarize the topic of the clusters using a recent approach [7]. Similar to the analysis of the keyword category, we show the utility impact on the different prompt topics (top-5 highest impacted and lowest impacted, separated by a green line) using the remove technique in Figure 5(c). When the remove technique generates a low impact on outputs, the prompt topic is usually not related to real-world knowledge. For example, in topic 3, the prompt topic is generating a movie script, which does not require real-world knowledge. Thus, hiding the city name, first name and last name did not have a high utility impact on the LLM outputs. On the other hand, when the remove technique generates a high impact on outputs, the prompt topic is usually related to real-world knowledge. For example, topic 9 is writing scripts about some events, and the keywords state, street, and county require real-world knowledge. Another example is the keyword “hickory glen mall” for the prompt “write a script about the history of hickory glen mall” where “hickory glen mall” is related to real-world knowledge. In summary, the utility impact relates to both keyword category and prompt topic, and our analysis reveals a detailed explanation of why the utility impact varies.

4) Utility inference performance: We found our utility inference model achieved Pearson and Spearman correlation values of 0.71 and 0.64 on the test set, respectively, demonstrating the effectiveness of our model. In addition, we show the confusion matrix of the main model output in Figure 6. Specifically, we group the inferences and labels into bins of 0.1 (e.g., (0, 1]), except in the case of < 0. We have the following observations: First, for each inferred value bin, the corresponding label (each column) generally follows the normal distribution. Thus, in most cases, the inferred utility impact will not diverge far from the label. Secondly, we define < 0.3 as positive (high utility impact) and ≥ 0.3 as negative (low or median impact), and we obtained a false positive rate of 0.49%. The low false positive rates imply the effectiveness of our LPPA system.

VI. Discussion

Training utility inference model using online LLM outputs.

When training the utility inference model, we used the Llama 2 7B model to generate outputs of the original and modified prompts. The similarity scores of the outputs are then used as labels to train the model. We use the Llama 2 7B model but not the online LLM model to save costs and time. However, in real applications, users will use the online LLM model, and the utility inference model needs to be trained with the online LLM outputs to enable it to predict online LLM utility impact. In addition, we assume that a local inference model can accurately infer the utility impact of the online LLM model. As we know, the online LLM model is much more powerful than local ones. Thus, using local LLMs to derive utility impact needs further investigation.

Versatility of prompts and sensitive information.

For simplicity, we consider 215K of 570K real user prompts, and our selection criteria is the first prompt of the conversation. However, we realize that these non-first prompts could also contain sensitive keywords and need protection. These non-first prompts rely more on the

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context, and the sensitive information may be related to multiple consecutive prompts. We consider that conversation prompts are more complicated due to the context. Thus, exploring privacy techniques in a conversation scenario requires additional research.

Inaccuracies of name entity recognition.

Through our analysis, we observed that the NER inaccuracies led to unnecessary analysis of wrong keywords, either not sensitive or not in the accurate category. To have inaccurate NER and the resulting modified prompts using inaccurate keywords would add noises to the training dataset, complicating the cases and negatively impacting the utility inference model. For example, “2px 5px rgba(0,0,0,0.1)” was recognized as a user agent category, “cloudflare.com” was recognized as an email category, and “145 + 529” was recognized as a phone number category. As these keywords may be the key information for the prompt, removing them may result in a significant change in the LLM output. Moreover, these useless prompts and keyword pairs would negatively affect usability.

VII. Conclusion

In this paper, we propose a local privacy-preserving prompt assistant (LPPA) to help users address the privacy concerns in the online LLM prompts. We first demonstrate the potential utility impact on the LLM output when hiding the privacy keywords using different privacy techniques, including remove, replace, mask, and rewrite. Then, we proposed a local utility inference model that estimates the impact of these modifications without exposing any data to external servers, thereby protecting user privacy. LPPA enables users to make wise decisions to balance between privacy and utility in online LLM usage. Through the evaluation of various privacy techniques, we have demonstrated that it is possible to significantly reduce the risk of exposing sensitive information while largely maintaining the utility of LLM responses. In summary, our LPPA not only addresses significant privacy issues associated with the use of online LLMs but also enhances user trust in these powerful technological tools. By continuously refining and adapting the LPPA, we can better protect user privacy while ensuring that the innovative potential of LLMs is not compromised by privacy concerns. The current work enables users to modify one keyword in their prompt, and our future work will address the multi-word cases.

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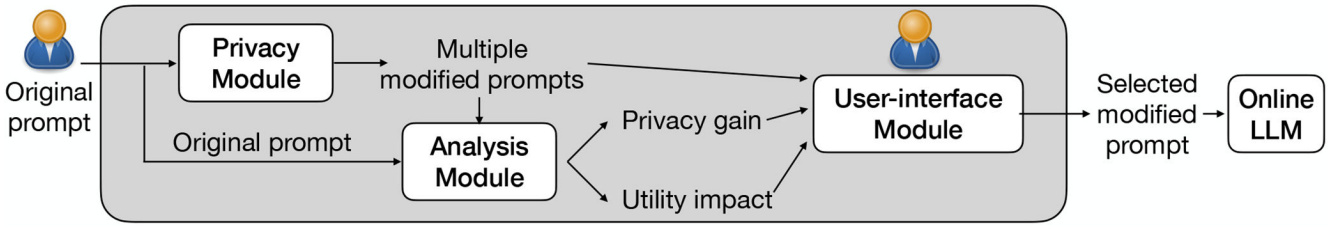


Fig. 1:
LPPA system

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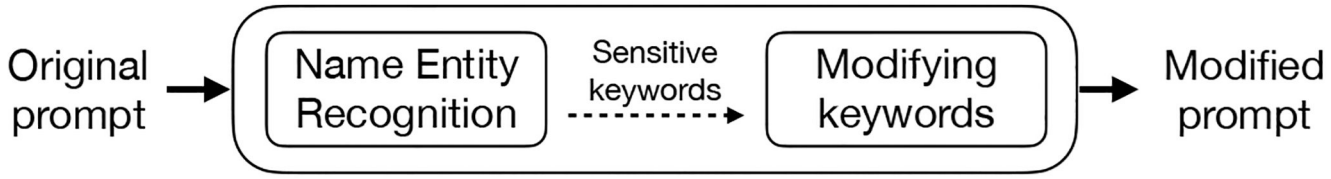


Fig. 2:
Privacy module

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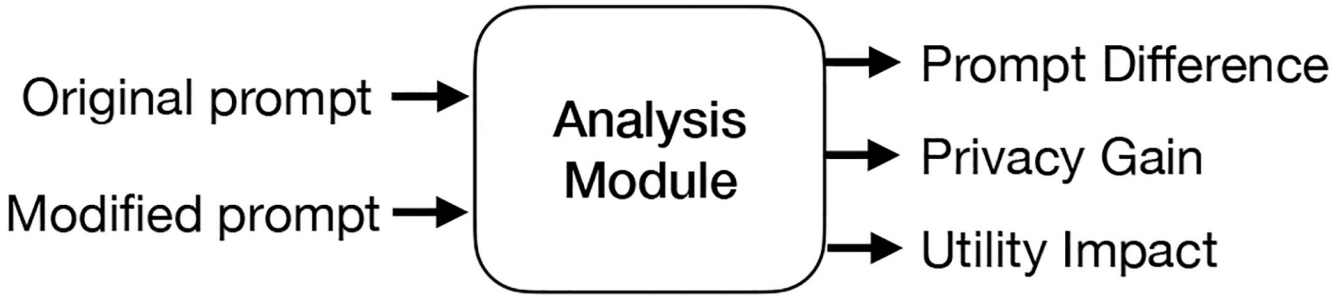


Fig. 3:
Analysis component

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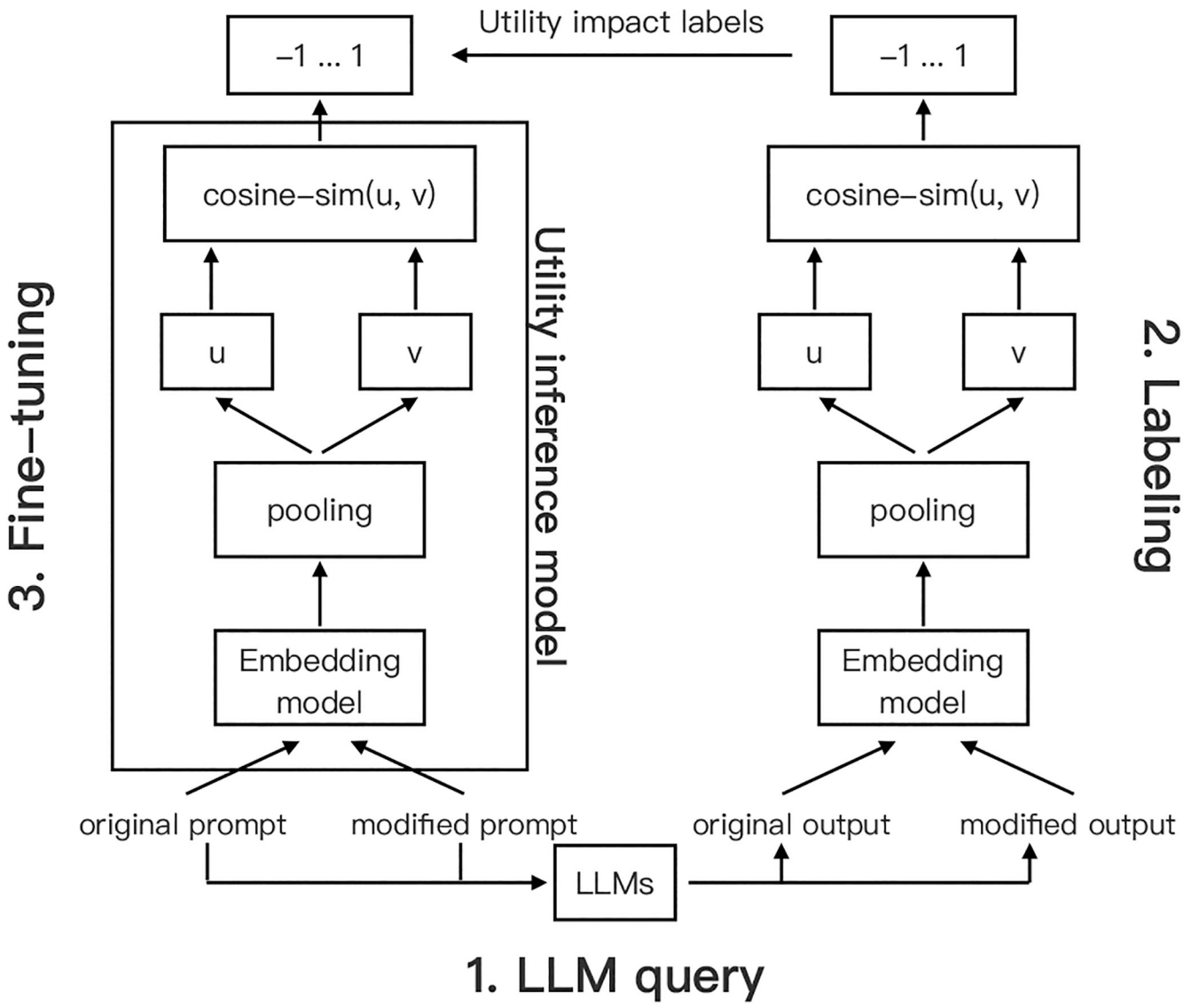
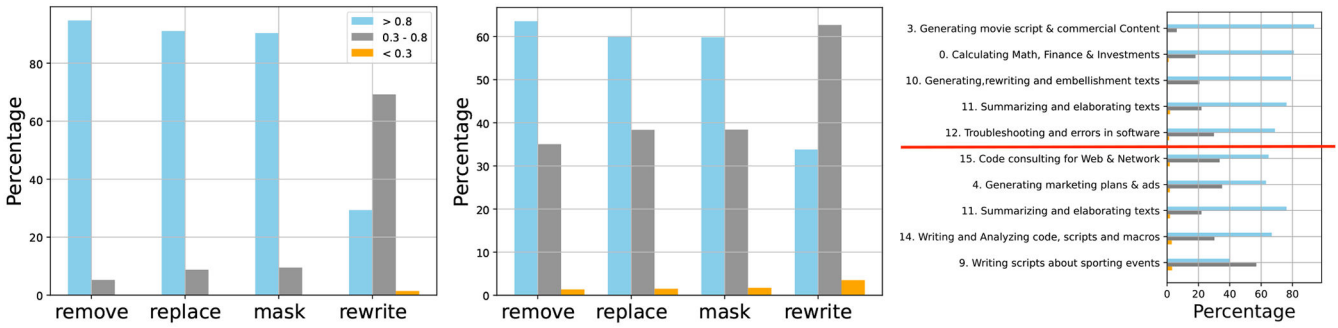


Fig. 4:
Utility inference model.



(a) Prompt similarity vs. techniques (b) Utility impact vs. techniques (c) Utility impact on vs. topics

Fig. 5:

Impact of privacy techniques on prompts and LLM outputs. We define semantic similarity (0.8, 1] as low impact, [0.3, 0.8] as median impact, and semantic similarity [−1, 0.3) as high impact.

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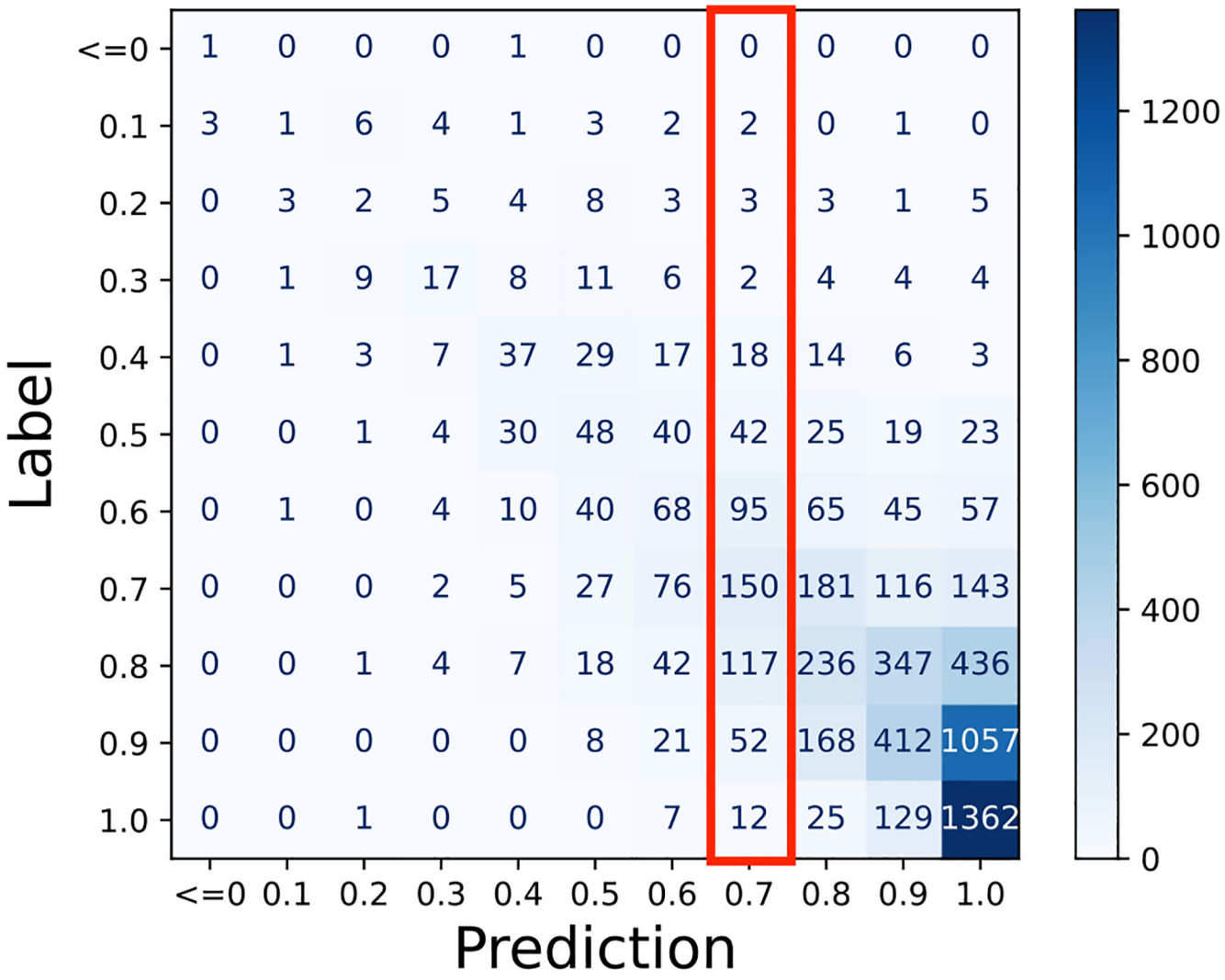


Fig. 6:
Confusion matrix of the utility impact prediction. The highlighted labels can be shown in the user-interface module.

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TABLE I:

Prompt difference. Prompt: “Write a short Seinfeld script where Kramer predicts 9/11, but Jerry, George, and Elaine dismiss him as a conspiracy theorist despite unknowingly getting all the facts of 9/11 right.”

		Similiarity			
Keywords	**Occurrence**	**Remove**	**Mask**	**Replace**	**Rewrite**
Kramer	1	0.98	0.93	0.95	0.74
Jerry, George	1	0.98	0.98	0.96	0.77
9/11	2	0.94	0.91	0.88	0.77